

Sampathkumar N

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Introduction

AI/ML systems engineer building research-grade computer vision, retrieval, and edge-AI systems, with four patents filed — two published at Indian-level conferences — and six independently engineered systems spanning agentic tooling, low-latency vector search, 3D vision, on-device voice AI, medical imaging, and neuromorphic robotics. Seeking a graduate research position applying systems-level ML engineering to open problems in efficient and embodied AI.

Education

Easwari Engineering College (SRM Group)

B.E., Computer Science and Engineering (Artificial Intelligence & Machine Learning)

• CGPA: 8.9 / 10

Chennai, India

Aug 2023 – May 2027

Patents & Publications

All patents filed through Easwari Engineering College (SRM Group); international filing in progress (~1 year timeline).

Knowledge Dark Matter Retrieval Network (KDM-RN) — Patent Filed; published at an Indian-level conference. Framework for inferring latent knowledge absent from existing knowledge graphs.

GeoMemory+: Terrain-Constrained Reconstruction of Historical Geography — Patent Filed; published at an Indian-level conference. Diffusion model reconstructing historical terrain from satellite and DEM data.

EduPulse AI: Personalized Learning & Career Guidance System — Patent Filed. Multilingual adaptive learning platform with an adaptive scoring engine. (Co-inventors: Duranjai V, Ilamurugan G.)

BaristaPOV VLM: Automated Analysis of Barista POV Video Streams — Patent Filed. Vision-language framework scoring beverage-preparation workflows. (Co-inventors: Duranjai V, Ilamurugan G.)

Research Projects

AutoPatch — Agentic AST-Based CI/CD Debugger and Autonomous Patcher

Python, AST, FastAPI

• Localizes test failures via AST traversal and synthesizes verified patch diffs — 100% pass@1 in 0.15s per fix.

HyperVector — Sub-Millisecond SIMD Vector Search Engine

C++20, ARM NEON, HNSW

• HNSW vector index with hand-written 128-bit NEON intrinsics — 21,750+ QPS at 0.046ms latency, 99.4% recall@10.

EdgeDepth-MPS — Real-Time Monocular-to-3D Vision on Apple Metal

PyTorch, MPS, Point Clouds

• Depth estimation with camera-intrinsic unprojection to dense point clouds — 46.8 FPS on-device.

EchoStream — Full-Duplex On-Device Voice AI

Apple MLX, WebSockets, VAD

• Streaming voice pipeline with barge-in handling — 23.66ms glass-to-glass latency, <40ms interrupt response.

NeuroScan-3D — Volumetric 3D MRI Brain Tumor Segmentation

PyTorch, 3D-UNet, BraTS

• Multi-modal 3D-UNet with multi-planar reconstruction — 94.1% Dice on whole tumor, 92.4% on tumor core.

NeuroBlink — Neuromorphic SNN for Drone Collision Evasion

SNN, Event Cameras, LIF Neurons

• Spiking neural network on event-camera data — 2.88ms evasion latency, 1,000x finer temporal resolution than frame cameras.

Experience

Machine Learning Intern, Tendworks Pvt Ltd

Mar 2026 – Present

Chennai, India

• Develop data-handling and workflow logic for the HRMS Expense Management module.

Machine Learning Intern, OEG Defence & Aerospace Pvt. Ltd.

Jun 2025

Chennai, India

• Built and deployed YOLOv8 drone object-detection models with Roboflow datasets and OpenCV inference pipelines.

Technical Skills

Languages: Python, C++, C, Java, SQL, JavaScript/TypeScript

AI/ML: PyTorch, Computer Vision, LLMs, RAG, Vision-Language Models, Spiking Neural Networks, Diffusion Models

Systems & Tools: Git, Docker, FastAPI, OpenCV, Roboflow, Vector Databases (pgvector, Redis), SQLite

Hardware Acceleration: ARM NEON SIMD, Apple Metal (MPS), Apple MLX

Honors & Activities

Top 30 of national teams, Smart India Hackathon 2024 (RailNav railway navigation system).

Participant: Convolve 4.0 (Pan-IIT AI/ML Hackathon), Dev Heat Hackathon (Spring Fiesta), Hack-4-Mini 2.0.

Certifications

Oracle Cloud Infrastructure 2025 Certified GenAI Professional — Oracle Cloud Infrastructure 2025 AI Foundations Associate

NPTEL: Internet of Things, Privacy & Security in Online Social Media, Leadership & Team Effectiveness

Infosys SpringBoard: Programming in C, Java Concepts, Introduction to Data Science